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Describe how a **non-competitive** inhibitor can reduce the rate of an enzyme-controlled reaction.

[3 marks]

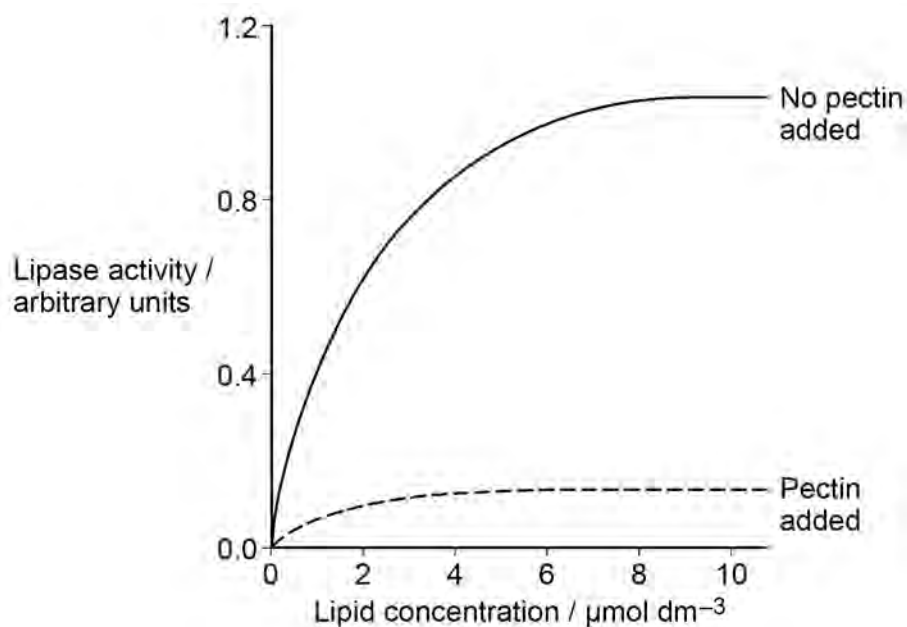
[illegible]

Pectin is a substance found in some fruit and vegetables.

A scientist investigated the effect of pectin on the hydrolysis of lipids by a lipase enzyme.

His results are shown in **Figure 1**.

Figure 1



0 1 2 The scientist concluded that pectin is a non-competitive inhibitor of the lipase enzyme.

Use **Figure 1** to explain why the scientist concluded that pectin is a **non-competitive** inhibitor.

[1 mark]

Question 1 continues on the next page

The scientist also found that pectin stops the action of bile salts. He prepared two suspensions:

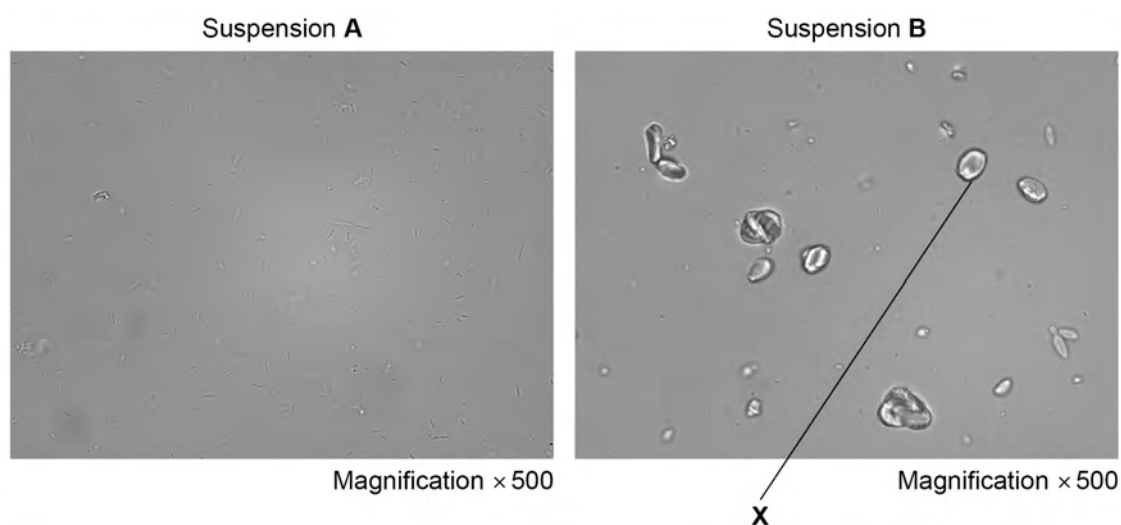
- suspension **A** – lipid and bile salts
- suspension **B** – lipid, bile salts and pectin.

He did **not** add lipase to either suspension.

He observed samples from the suspensions using an optical microscope.

Figure 2 shows what he saw in a typical sample from each suspension.

Figure 2



0 1 . 3 Calculate the maximum length of the large lipid droplet marked **X** in **Figure 2**.

Using a ruler with millimetre intervals always includes an uncertainty in the measurement. Use the uncertainty in your measurement to determine the uncertainty of your calculated maximum length.

You can assume there is no uncertainty in the magnification.

[2 marks]

Maximum length = _____ μm

Uncertainty of your calculated maximum length = _____ μm

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No large lipid droplets are visible with the optical microscope in the samples from suspension **A**.

Explain why.

[2 marks]

8

Turn over for the next question

Turn over ►

